

Robot Kinematics and Vision (RKV)

Term Project – Feed-Forward 3D Reconstruction
(MapAnything)

Project Resources

MapAnything project page: <https://map-anything.github.io/>

MapAnything code: <https://github.com/facebookresearch/map-anything>

COLMAP & MapAnything overview video: <https://www.youtube.com/watch?v=z4L9NLHKScM>

Build & run tutorial video: <https://youtu.be/Z8kJ86JdxB8>

AR capture tutorial video: <https://www.youtube.com/watch?v=H3ssOq7W-ho>

- COLMAP and MapAnything are run on **Google Colab** or **Elice cloud GPU**. See the tutorial videos above for build & run instructions.
- **Tasks 1–2 are required. Task 3 (ARKit / ARCore) is optional** and may be replaced if device capture is not feasible.
- **Questions:** please post on the **LMS Q&A board** (not via personal email to the TAs). TAs will respond there in order.

Evaluation Criteria

- Whether COLMAP and MapAnything training and rendering are successful on the RGB image sequence
- Understand the camera frames and inertial coordinate used in COLMAP and MapAnything
- Comparison between the actual image (true) and rendered (estimated) image
- Quantitative comparison across input configurations (uncalibrated vs. calibrated, with vs. without poses)

Submission & Schedule

Final presentation: Wed, June 10 — ~10 min per team, in class

Final report deadline: Fri, June 19, 23:59 (KST)

[Final Report Must Include]

- **The name of your RGB image sequence**

Briefly explain which RGB image sequence you used and why.

- **Rendered results from multiple viewpoints**

Show the rendered results from each method (COLMAP, MapAnything), captured from various viewpoints.

- **Analysis of the results**

Provide your own analysis:

Was the result good or bad? According to what criteria?

How should the results be interpreted or evaluated?

Include any insights, issues, or observations you found.

Project Tasks & MapAnything Background

- **Task 1.** Run **COLMAP** on your RGB image sequence and obtain camera intrinsics + poses
- **Task 2.** Run **MapAnything** under 4 input configurations (uncalibrated/calibrated, w/wo poses)
- **Task 3.** **ARKit / ARCore + MapAnything** (optional)

MapAnything paper

Keetha, Müller, Schönberger, et al.

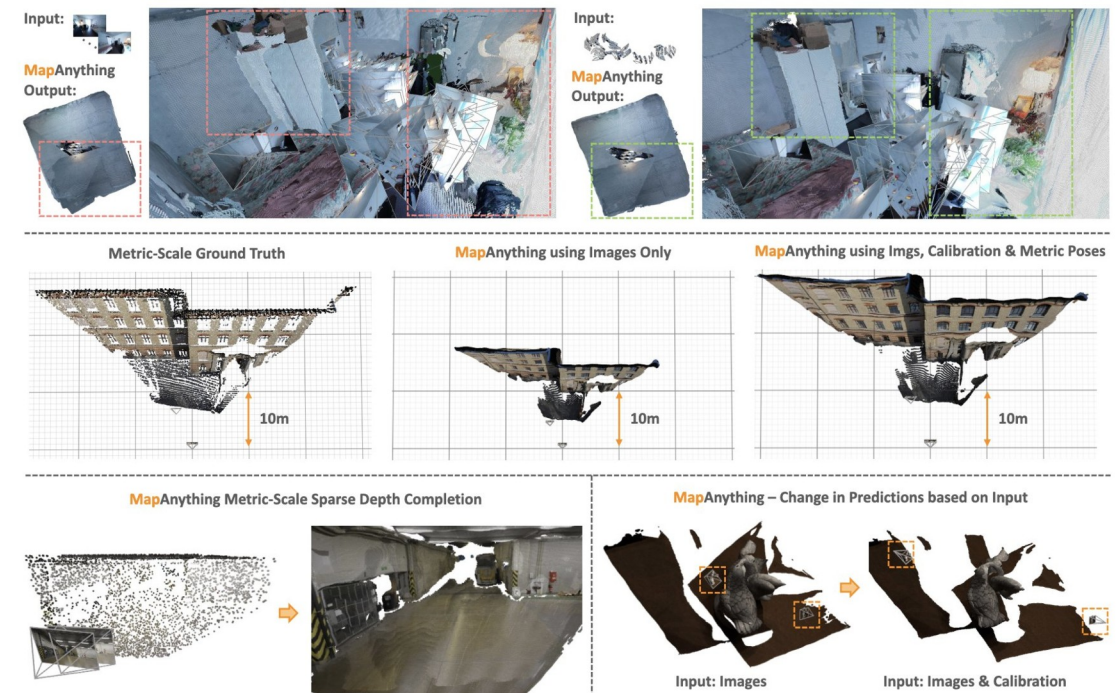
MapAnything: Universal Feed-Forward Metric 3D Reconstruction.

International Conference on 3D Vision (3DV), IEEE, 2026.

<https://arxiv.org/abs/2509.13414>

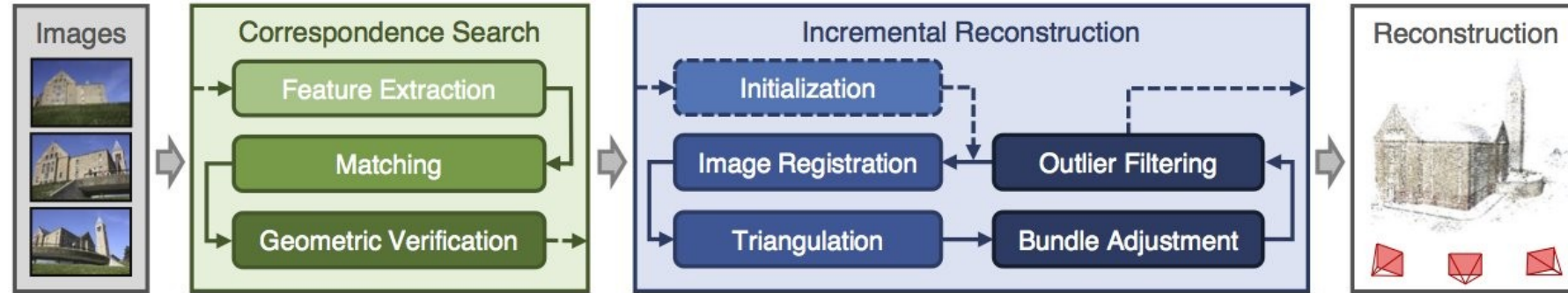
Key idea: a unified **transformer-based feed-forward** model that optionally accepts extra geometric inputs (intrinsics, poses, depth) — exactly the 4 configurations of Task 2.

See also: [COLMAP & MapAnything overview video](#)



Task 1. COLMAP – Classical SfM

Goal: obtain camera intrinsics and poses that will be used as the reference in Task 2.



- 1) Build & run COLMAP on Google Colab or Elice (see tutorial video)
- 2) Use your own RGB image sequence as input
- 3) Run feature extraction -> matching -> incremental mapping
- 4) Export the COLMAP output: cameras.txt (intrinsics), images.txt (poses), points3D.txt (3D points)
- 5) Visualize the reconstructed point cloud and camera trajectory
- 6) Submit: number of registered images, mean reprojection error, screenshot of the reconstruction

References

- Build & run tutorial video: <https://youtu.be/Z8kJ86JdxB8>
- Official COLMAP tutorial: <https://colmap.github.io/tutorial.html>

Task 2. MapAnything – Feed-Forward 3D

Goal: run MapAnything on Colab/Elice GPU under 4 input configurations and quantify how the input affects the output.

[Run MapAnything 4 times — one per configuration]

- **Config A. Uncalibrated, image-only** – feed raw RGB images. No intrinsics, no poses.
- **Config B. Calibrated, image-only** – feed raw RGB images + intrinsics from Task 1. No poses.
- **Config C. Calibrated + COLMAP poses** – feed RGB images + intrinsics + Task 1 COLMAP poses.
- **Config D. Calibrated + AR poses** – Task 3 (optional): feed images + intrinsics + ARKit/ARCore poses.

[For each configuration, render novel views and report PSNR / SSIM, predicted pose error, runtime, etc]

Comparisons: A vs. B (effect of calibration) / **B vs. C** (effect of poses) / **(optional) C vs. D** (pose source: COLMAP vs. AR)

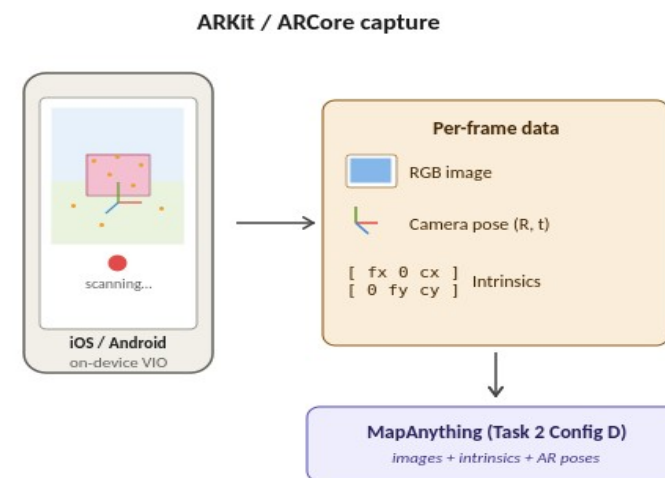
References

- Build & run tutorial video: <https://youtu.be/Z8kJ86JdxB8>
- MapAnything project page / code: <https://map-everything.github.io/>

Task 3. ARKit / ARCore + MapAnything (optional)

Goal: bypass COLMAP entirely by feeding on-device VIO poses directly into MapAnything.

- 1) Capture a short scan of a static scene with ARKit (iOS) or ARCore (Android). [\[AR capture tutorial video\]](#)
- 2) Convert the saved AR poses into the MapAnything coordinate convention; document the conversion
- 3) Run MapAnything on the AR-posed images — this is Task 2 Config D
- 4) Compare against Task 2 Config C (which used COLMAP poses) on the SAME scene: report PSNR / SSIM, pose drift, runtime, etc



References

- AR capture tutorial video: <https://www.youtube.com/watch?v=H3ssOq7W-ho>
- ARKit documentation: <https://developer.apple.com/documentation/arkit>
- ARCore documentation: <https://developers.google.com/ar/develop>

Useful References

- COLMAP & MapAnything overview video: <https://www.youtube.com/watch?v=z4L9NLHKScM>
- Build & run tutorial video: <https://youtu.be/Z8kJ86JdxB8>
- AR capture tutorial video: <https://www.youtube.com/watch?v=H3ssOq7W-ho>
- MapAnything paper (3DV 2026 / arXiv:2509.13414): <https://arxiv.org/abs/2509.13414>
- MapAnything project page: <https://map-anything.github.io/>
- MapAnything code: <https://github.com/facebookresearch/map-anything>
- Official COLMAP tutorial: <https://colmap.github.io/tutorial.html>
- COLMAP detailed description (CVPR 2017): <https://demuc.de/tutorials/cvpr2017/sparse-modeling.pdf>
- Background: NeRF: <https://github.com/bmild/nerf>
- Background: DUST3R: <https://github.com/naver/dust3r>
- Background: Official 3D GS source code: <https://github.com/graphdeco-inria/gaussian-splatting>